

A student team is developing a CanSat for a high-altitude atmospheric mission.

The CanSat will be launched aboard a sounding rocket to an altitude of approximately **1000 m** above ground level.

After deployment from the rocket, the CanSat must descend safely while collecting and transmitting scientific and engineering data.

Your task is to design the complete Electrical Subsystem and select the necessary avionics and sensors required to fulfill the mission objectives.

The final design should be realistic, technically feasible, and based on commercially available components.

Mission Requirements

The CanSat shall:

R1: Measure atmospheric pressure continuously throughout the mission.

R2: Measure ambient temperature continuously throughout the mission.

R3: Measure acceleration and orientation of the CanSat.

R4: Determine and transmit GPS/GNSS position during flight.

R5: Measure and report battery voltage and battery current.

R6: Transmit telemetry to a ground station at a minimum rate of 1 Hz.

R7: Store all sensor data locally in case telemetry is lost.

R8: Record video from deployment until landing.

R9: Provide an audio recovery beacon after landing.

R10: Operate continuously for:

- 2 hours on the launch pad
- 15 minutes of flight operations

with a minimum design margin of 25%.

Design Constraints

C1: Total CanSat mass shall not exceed 1.0 kg

C2: The Electrical Subsystem mass shall not exceed 300 g

C3: Lithium-polymer batteries are strictly prohibited

C5: Maximum project budget for electrical components: ₹10,000

C6: Only commercially available components may be used.

C7: The design must survive:

- launch vibration
- shock loading
- temporary communication loss

Required Deliverables

Part A — System Architecture

Create a block diagram showing:

- Battery
- Voltage regulators
- Microcontroller
- Sensors
- Telemetry module
- Data storage
- Camera
- Recovery beacon

All power and communication connections should be clearly shown. Here, we don't need the exact wiring, individual resistors, capacitors etc. We just need you to show how these components are connected conceptually.

Part B — Component Selection

Select components for:

Pressure Sensor

Temperature Sensor

IMU

GNSS Module

Microcontroller

Telemetry Module

Data Storage

Camera

Battery Monitoring System

Recovery Beacon

For each selected component provide:

Parameter	Value
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Component Name	
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Operating Voltage	
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Current Consumption	
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Communication Interface	
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Cost	
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Justification	
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Part C — Trade Study

For at least:

- one sensor category
- the microcontroller
- the telemetry module

compare a minimum of:

3 candidate components

and justify the final selection.

Example:

Component	Advantages	Disadvantages	Score
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Part D — Electrical Power Subsystem Design

Design the complete EPS.

Include:

Battery Selection

Determine:

- battery type
 - configuration
 - capacity
 - total stored energy
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Power Distribution

Specify:

- voltage rails
- regulator types
- protection mechanisms

Example:

Rail	Voltage	Loads
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Power Protection

Discuss:

- reverse polarity protection
 - over-current protection
 - switch design
 - battery mounting strategy
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Part E — Power Budget

Create a detailed power budget.

Example format:

Component	Voltage	Current	Power
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Calculate:

Peak Power Consumption

Average Power Consumption

Total Mission Energy Consumption

Required Battery Capacity

Battery Margin

All calculations must be shown.

Part F — Circuit Design (Electrical Schematic)

Create a schematic showing:

Battery

Regulators

MCU

Sensors

GNSS

Telemetry Module

Data Storage

Camera

Beacon

Here, you have to show how the components would actually be connected. Create a circuit-level schematic showing how the selected components are electrically connected, including power rails, communication interfaces, and essential supporting circuitry.

The schematic may be created using:

- KiCad
- EasyEDA
- Altium

- Proteus
- Draw.io
- hand-drawn scanned diagram

provided it is legible.